40.520 Stochastic Models
Comprehensive Practice Paper (50 Questions)

Part I: Examination Paper (50 Questions)

Answer all questions. Justify clearly.

Section A: Probability Foundations (Q1–Q20)

1. State the three Kolmogorov axioms.
2. Prove that $P(A^c) = 1 - P(A)$.
3. Prove inclusion-exclusion for two events.
4. Define conditional probability and derive Bayes' theorem.
5. State and prove the Law of Total Probability.
6. Define independence. Show that independence implies $P(A|B) = P(A)$.
7. Let X be discrete with pmf $p(x)$. Define $E[X]$ and prove linearity.
8. Show that $Var(X) = E[X^2] - (E[X])^2$.
9. If X, Y independent, prove $Var(X + Y) = Var(X) + Var(Y)$.
10. Let $X \sim Bern(p)$. Compute mean and variance.
11. Define joint pmf and marginal pmf.
12. Define conditional expectation $E[X|Y]$.
13. State and prove the Law of Iterated Expectations.
14. Prove Markov's inequality.
15. Prove Chebyshev's inequality.
16. State the Weak Law of Large Numbers.

17. Conceptually distinguish WLLN and SLLN.
18. State Central Limit Theorem.
19. Define convergence in probability.
20. Define time average vs ensemble average.

Section B: Discrete Time Markov Chains (Q21–Q50)

21. Define a discrete-time Markov chain.
22. State the Markov property formally.
23. Show that $P(X_4|X_3, X_2, X_1) = P(X_4|X_3)$.
24. Define transition matrix.
25. Show rows of transition matrix sum to 1.
26. Define n-step transition probability.
27. State and prove Chapman-Kolmogorov.
28. Define communication of states.
29. Define irreducibility.
30. Define recurrence and transience.
31. Define positive vs null recurrence.
32. Define periodicity.
33. Define stationary distribution.
34. Show $\pi = \pi P$ condition.
35. Define limiting probability.
36. State Ergodic Theorem.
37. Define hitting time.
38. Define hitting probability.
39. Define stopping time.
40. Solve gambler's ruin with $p \neq q$.
41. Show infinite gambler ruin is certain if $p \leq 1/2$.

42. For 2-state chain, compute stationary distribution.
43. For birth-death chain derive detailed balance.
44. Show irreducible finite chain has stationary distribution.
45. Explain difference between stationary and limiting distribution.
46. Show if chain is irreducible and positive recurrent then $\lim P_{ij}^n = \pi_j$.
47. Show future and past are conditionally independent given present.
48. Compute expected hitting time for simple symmetric random walk on 0,...,N.
49. Define absorbing state.
50. Show absorbing state is recurrent.

Part II: Model Answers (Concise)

1. Non-negativity, normalization, countable additivity.
2. From $A \cup A^c = \Omega$ and disjointness.
3. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
4. $P(A|B) = \frac{P(A \cap B)}{P(B)}$, derive Bayes.
5. Partition argument.
6. Independence definition.
7. $E[X] = \sum xp(x)$, linearity via sum properties.
8. Expand $(X - E[X])^2$.
9. Expand variance.
10. $E[X] = p$, $Var = p(1-p)$.
11. Sum over other variable.
12. $E[X|Y] = \sum xP(X = x|Y)$.
13. $E[E[X|Y]] = E[X]$.
14. Use indicator trick.
15. Apply Markov to $(X - \mu)^2$.
16. $P(|\bar{X}_n - \mu| > \epsilon) \to 0$.
17. Almost sure stronger.
18. Normal limit.
19. Definition via probability limit.
20. Sample path vs expectation.
21. Markov chain definition.
22. Conditional independence of future given present.
23. Direct from Markov property.
24. Matrix of one-step probabilities.
25. By total probability.
26. $P_{ij}^{(n)}$ definition.

27. Sum over intermediate states.
28. $i \leftrightarrow j$ if accessible both ways.
29. Single class chain.
30. Recurrence definition.
31. Finite mean return time.
32. gcd of return times.
33. $\pi_j = \sum_i \pi_i p_{ij}$.
34. Definition.
35. $\lim_{n\to\infty} P_{ij}^n$.
36. Equivalence of long-run, stationary, limiting.
37. $T = \inf\{n : X_n \in A\}$.
38. $h_i = P_i(T < \infty)$.
39. Measurable wrt history.
40. $h_i = \frac{1-(q/p)^i}{1-(q/p)^N}$.
41. Martingale argument.
42. Solve linear system.
43. $\pi_i p_{ij} = \pi_j p_{ji}$.
44. Perron-Frobenius.
45. Stationary need not imply convergence.
46. From ergodicity.
47. From Markov property factorization.
48. Solve second-order difference equation.
49. State that never leaves.
50. Return probability =1.